

ISLAMIC UNIVERSITY OF TECHNOLOGY(IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)
CSE 4603: Software Testing & Quality Assurance
Quiz 4 SET A

Time 15 minutes

Marks 20

Student ID : _____

1. Which activity does not require software measurements? 1
 - a. Development
 - b. Understanding
 - c. Control
 - d. Improvement
2. Which of the followings can be called a primitive metrics? 1
 - i. LOC
 - ii. FPA
 - iii. Defect Density
 - a. i only
 - b. i & iii
 - c. ii & iii
 - d. i, ii & iii
3. Which of the following is a example of external attribute? 1
 - a. Total test points
 - b. Defect Density
 - c. Productivity
 - d. Program level
4. Mention two problems with using LOC as a comparable metric. 2

__Program Language dependent_____

__Some lines are different than others_____
5. In a module implementation of a project, there are 17 unique operators, 12 unique operands, 45 total operators, and 32 total operands appearing in the module implementation. Calculate its Halstead Program Level. 2

$PL = 1 / ((17/2) * (32/12)) = 0.0441$

_____0.04411_____

6. Why is identifying boundaries important for calculating function points? 1
- Including functions in the calculation
 - Identifying the user and external source
 - Define the scope of the project
 - All of the above

7. Mention each of the function types? Using any of these types - Internal Logical Files, External Interface Files, External Input, External Output and External Inquiries. 5

Scrolling _____ Inquires _____

Confirmation Message on clicking submit _____ Output _____

Store Customer information _____ Internal Logical Files _____

Error Messages _____ Output _____

Notification Messages _____ Output _____

8. Out of all 14 general characteristics of a system, ABC project only requires five of them (Data communication, Transaction rate, Performance, Online data entry and Online update) at a significant (4) level. Calculate the value adjusted factor for the mentioned project. 2

$$\text{____ VAF} = 5 \times 4 \times 0.01 + 0.65 = 0.85 \text{_____}$$

9. From a project, function point was calculated at 460 and the number of bugs found was 103. Calculate the defect density of this project. 1

$$\text{____ } 103/460 = 0.22 \text{_____}$$

10. For a project, function point was calculated at 320, and the values for function dependent factors taken into account are: Function Importance - High (12), Use intensity - Normal (4), Interfacing - High (8), Complexity - Normal (4). Consider all dynamic characteristics as very important (5). Calculate the dynamic test point for this project. 4
- FP=320;

$$\text{FDC}_w = ((12+4+8+4)/20) \times 1 = 1.4$$

$$\text{QC}_{dw} = (5 \times 0.75 + 5 \times 0.05 + 5 \times 1 + 5 \times 0.1) / 4 = 5/4 = 1.25$$

$$\text{____ } 320 \times 1.4 \times 1.25 = 560 \text{_____}$$